

What is the Component Lifecycle?

Every React component goes through three phases:

1. Mounting

The component is created and inserted into the DOM.

Methods:

- `constructor()`
- `render()`
- `componentDidMount()`

2. Updating

Occurs whenever state or props change.

Methods include:

- `shouldComponentUpdate()`
- `componentDidUpdate()`

3. Unmounting

Component is removed from the DOM.

Method:

- `componentWillUnmount()`

Why use Lifecycle Methods?

- Fetch data from APIs
- Initialize state
- Update UI
- Handle errors
- Release resources

`componentDidMount()`

Called **once** after the component is rendered.

Ideal for:

- Fetch API
- Database calls
- Timers

Example:

```
componentDidMount() {  
  this.loadPosts();  
}
```

componentDidCatch()

Acts as an Error Boundary.

Called when a child component throws an error.

Used to:

- Display alerts
- Log errors
- Prevent application crashes

Practical:

Post.js:

```
class Post {  
  constructor(userId, id, title, body) {  
    this.userId = userId;  
    this.id = id;  
    this.title = title;  
    this.body = body;  
  }  
}
```

```
}
```

```
export default Post;
```

Posts.js:

```
import React, { Component } from "react";
```

```
import Post from "./Post";
```

```
class Posts extends Component {
```

```
  constructor(props) {
```

```
    super(props);
```

```
    this.state = {
```

```
      posts: []
```

```
    };
```

```
  }
```

```
  loadPosts() {
```

```
    fetch("https://jsonplaceholder.typicode.com/posts")
```

```
      .then(response => response.json())
```

```
      .then(data => {
```

```
        const posts = data.map(item =>
```

```
          new Post(
```

```
            item.userId,
```

```
            item.id,
```

```
        item.title,
        item.body
    )
);

this.setState({
    posts: posts
});

})
.catch(error => {
    throw error;
});
}

componentDidMount() {
    this.loadPosts();
}

componentDidCatch(error, info) {
    alert("Error: " + error.message);
    console.log(info);
}

render() {
```

```
return (  
  
  <div>  
  
    <h1>Blog Posts</h1>  
  
    {this.state.posts.map(post => (  
  
      <div key={post.id}>  
  
        <h2>{post.title}</h2>  
  
        <p>{post.body}</p>  
  
        <hr />  
  
      </div>  
  
    )})  
  
  </div>  
  
  );  
}
```

```
}
```

```
export default Posts;
```

App.js:

```
import React from "react";
```

```
import Posts from "../Posts";
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
      <Posts />
```

```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```

Output:

